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Monsoon, 2024

B. Tech 3rd Semester Examination

Civil Engineering, Materials, Testing & Evaluation

Course Code: BCE23302T

Full Marks – 50

Time – 2 hours

The figure in the margin indicates full marks for the questions.

Answer all the questions

Marks: 10 x 1 = 10

- I. (a) What is the main purpose of adding gypsum to cement?
 - i. To increase strength
 - ii. To reduce setting time
 - iii. To prevent cracking
 - iv. To control the setting time
- (b) What is the main disadvantage of using timber as a building material in construction?
 - i. High cost
 - ii. Susceptibility to fire and decay
 - iii. Poor aesthetic appearance
 - iv. Low strength-to-weight ratio
- (c) What is the primary function of aggregates in concrete?
 - i. To improve the thermal insulation properties
 - ii. To reduce the cost of concrete
 - iii. To enhance the strength and durability of concrete
 - iv. To prevent the concrete from cracking
- (d) In the context of elasticity, Hooke's Law states that stress is proportional to strain up to the:
 - i. Yield point
 - ii. Ultimate tensile strength
 - iii. Elastic limit
 - iv. Fracture point
- (e) Which of the following is NOT a characteristic of a material that exhibits elastic behavior?
 - i. It returns to its original shape after the load is removed.
 - ii. The material deforms permanently after the load is removed.
 - iii. It obeys Hooke's Law within its elastic limit.
 - iv. The material undergoes linear deformation under stress.

(f) Poisson's ratio is defined as the ratio of:

- ☒ i. Lateral strain to longitudinal strain
- ii. Longitudinal strain to lateral strain
- iii. Shear stress to shear strain
- iv. Longitudinal stress to lateral stress

(g) In a bending test, the material of the specimen is subjected to:

- i. Shear stress only
- ☒ ii. Compressive and tensile stress
- iii. Pure bending stress
- iv. Axial stress

(h) The ability of a material to resist deformation under a twisting load is known as:

- i. Shear modulus
- ☒ ii. Torsional strength
- iii. Bending strength
- iv. Compressive strength

(i) The primary purpose of the impact test is to measure a material's:

- i. Hardness
- ii. Ductility
- ☒ iii. Toughness
- iv. Yield strength

(j) Which of the following is the most common impact test used to measure a material's resistance to sudden fractures?

- ☒ i. Charpy Test
- ii. Brinell Test
- iii. Rockwell Test
- iv. Vickers Test

- 2(a) Discuss the various types of cements used in construction and their properties. How do different types of cement affect the strength and durability of concrete? (6+4=10)

OR

- 2(b) What are the differences between paints and varnishes? Describe the applications of each in construction and finishing. (6+4=10)

Discuss the various functions of Bogue's Compound.

- 3(a) Draw the stress-strain curve for mild steel and explain the different regions observed in the curve. Discuss the key points such as elastic limit, yield point, ultimate tensile strength, and breaking stress. (5+5=10)

OR

- 3(b) Explain the difference in tensile testing procedures and standards for brittle materials compared to ductile materials. (10)

- 4 (a) Explain the bending and torsion tests used to determine the material strength of a specimen. For each test, describe the procedure, the expected behavior of a material under these loads, and the key factors influencing the material's strength. (4+4+2=10)

OR

- 4 (b) Define creep in materials and explain the stages of creep deformation (primary, secondary, and tertiary). What are the factors affecting creep. (3+4+3=10)

- 5 (a) Discuss the concepts of elastic and plastic deformation in materials. Explain how these types of deformation are characterized and how they influence the mechanical properties of materials. (5+5=10)

OR

- 5(b) Describe the various types of mechanical tests commonly conducted in a material testing laboratory and explain the purpose of each test. (5+5=10)